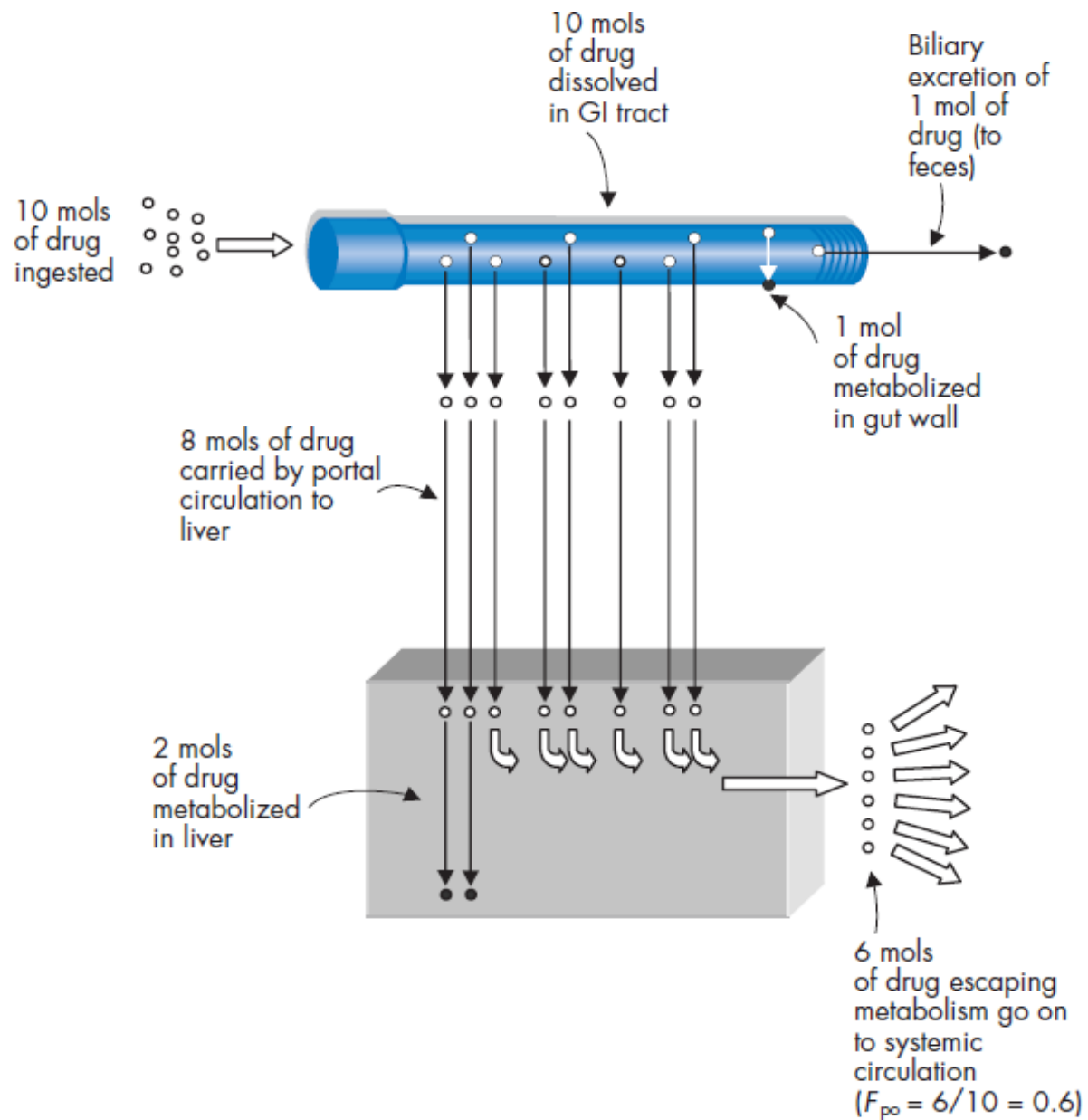


2.7 PK of Extravascular Administration

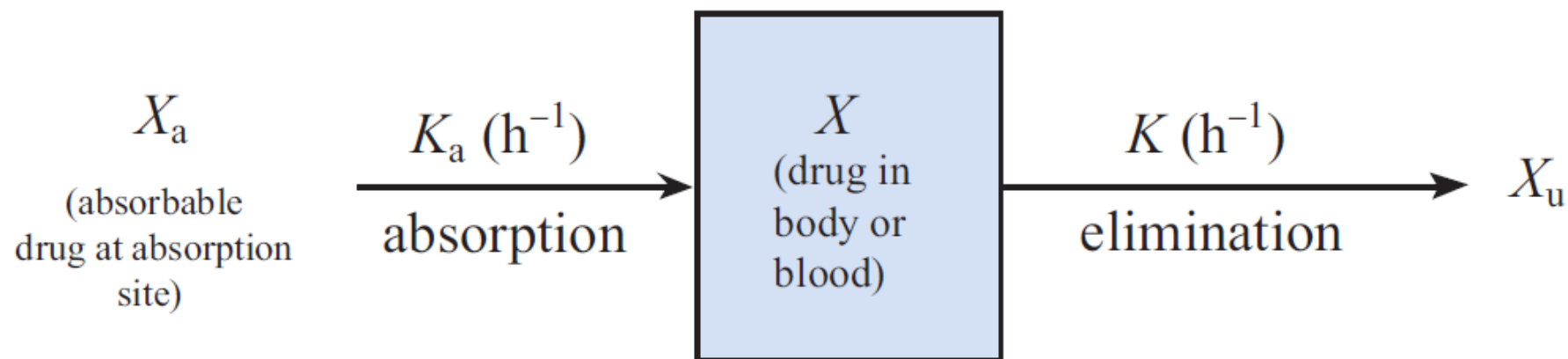


- For extravascular administration, absorption is a prerequisite for pharmacological effects.
- Delays or drug loss during absorption may contribute to variability in drug response and, occasionally, may result in a failure of drug therapy.

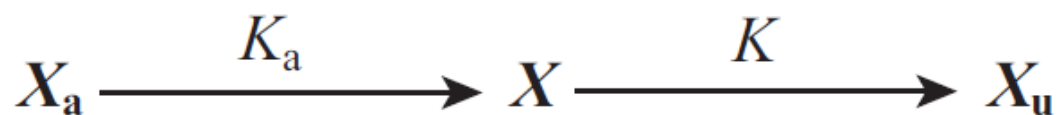




- PK parameters, from C_p Vs time data, following oral administration of drugs, are determined with the following **assumptions**:
 - Drug exhibits the characteristics of one compartment model,
 - Absorption and elimination of a drug follow the first-order process and passive diffusion,
 - Drug is eliminated in unchanged form (i.e. no metabolism occurs),
 - Drug is monitored in the blood.



SETUP:





- For a drug that enters the body by an apparent first-order absorption process, is eliminated by a first-order process, and distributes in the body according to a one-compartment model, the following differential equation applies:

$$\frac{dX}{dt} = K_a X_a - KX$$

where X and K are as defined previously, K_a is the apparent first order absorption rate constant, and X_a is the amount of drug at the absorption site.

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- Assuming 'F' is the fraction of the oral dose (X_o) that will be absorbed ultimately;
- Then, the amount of drug available for absorption at $t = 0$,
 X_a^0

$$X_a^0 = FX_o$$



- The rate of disappearance of the drug from the absorption site is described by:

$$\frac{dX_a}{dt} = -K_a X_a \quad \text{or} \quad X_a = X_a^0 e^{-K_a t}$$

- Substituting $F X_0$ for X_a^0

$$X_a = F X_0 e^{-K_a t}$$



- Substituting $FX_0e^{-K_at}$ for X_a in the differential equation for the amount of the drug in body ($\frac{dX}{dt} = K_aX_a - KX$):

$$\frac{dX}{dt} = K_aFX_0e^{-K_at} - KX$$

- Integrating this equation:

$$X = \frac{K_aFX_0}{(K_a - K)} (e^{-Kt} - e^{-K_at})$$

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- Dividing the above equation by V yields:

$$\frac{X}{V} = C = \frac{K_a F X_o}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$



Elimination half life ($t_{1/2}$) and elimination rate constant (K)

- For most drugs administered extravascularly in conventional dosage forms, $K_a \gg K$.
- As a result, at some time after administration, the term $e^{-K_a t}$ approaches zero, whereas the term e^{-Kt} is finite,
- Therefore, the eqn. $\left[C = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t}) \right]$ reduces to

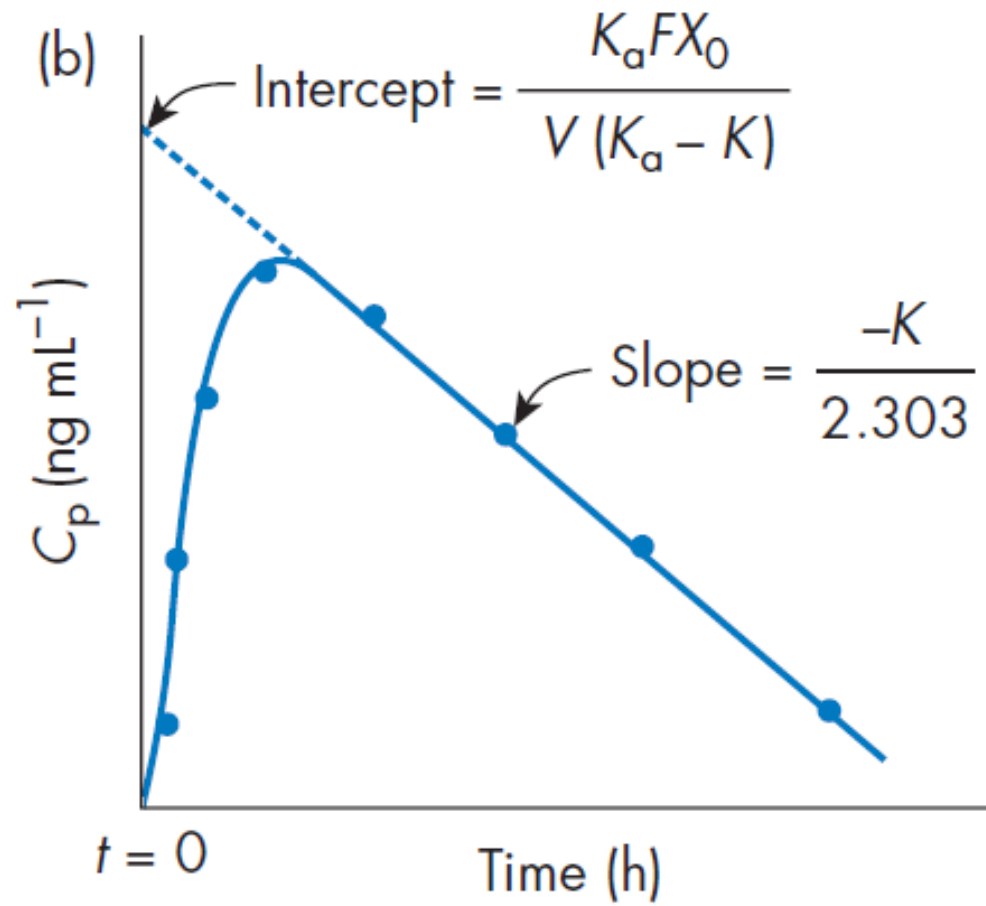
$$C = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt})$$



- A plot of the logarithm of drug concentration in plasma versus time yields a biexponential curve, with a terminal linear portion having a slope of $-\frac{K}{2.303}$.

$$\log C = \log \left[\frac{K_a F X_0}{V(K_a - K)} \right] - \frac{Kt}{2.303}$$

- Elimination half-life of the drug is given by $\frac{0.693}{K}$.





Absorption rate constant (K_a)

- The absorption rate constant is determined by a method known as **“method of residuals”**
- The method allows the separation of the monoexponential constituents of a biexponential plot of plasma concentration against time.
- This graphical approach for estimating K_a is useful only if K_a and K rate constants are substantially different.
 - In general, the method works best if $\frac{K_a}{K} \geq 3$.



- If this is not the case (i.e. if $\frac{K_a}{K}$ is not ≥ 3), the rate constants are best estimated by fitting the concentration-time data to:

$$C = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

with the aid of a nonlinear least-squares regression program.



Method of residuals

- From the equation,

$$C = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

- Assuming that $K_a \gg K$, the term $e^{-K_a t}$ will approach zero, whereas the term e^{-Kt} retains a finite value, then the equation reduces to:

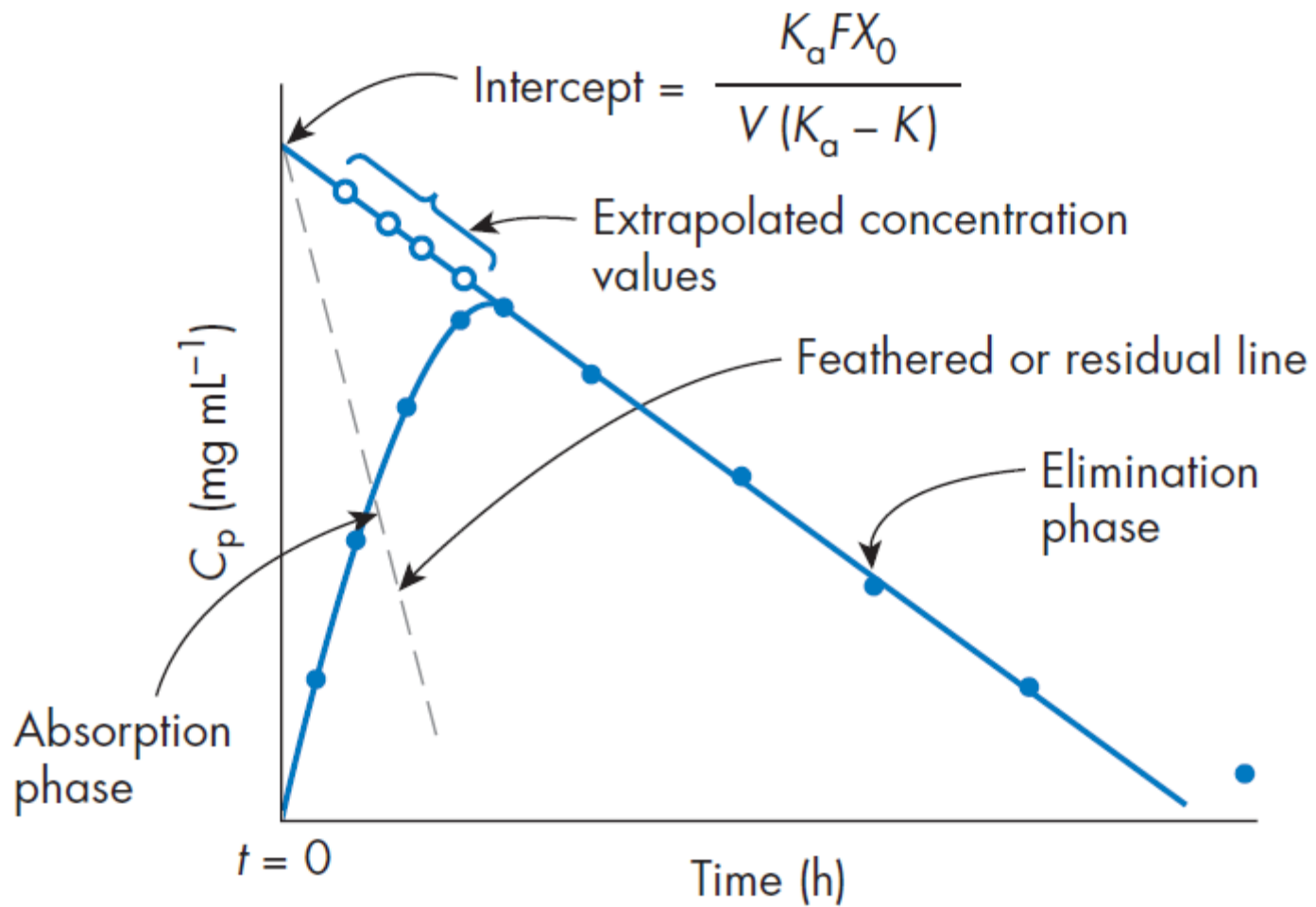
$$C = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt})$$



- A plot of the $\log C$ vs. t yields a biexponential curve
 - Extrapolation of the terminal linear phase to time zero will yield an intercept equal to:

$$\log \left[\frac{K_a F X_0}{V(K_a - K)} \right]$$

- Subtraction of the true plasma concentration values in the absorptive phase from the corresponding concentration values on the extrapolated line yields a **series of residual concentrations**.



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- These residual values are described by:

$$C_R = C_{extrapolated} - C_{observed}$$

And,

$$C_{extrapolated} = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt})$$

$$C_{observed} = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$



- Therefore,

$$C_R = \frac{K_a F X_0}{V(K_a - K)} (e^{-K_a t})$$

- In terms of common logarithms

$$\log C_R = \log \left[\frac{K_a F X_0}{V(K_a - K)} \right] - \frac{K_a t}{2.303}$$

- Then, a plot of $\log C_R$ vs t yield a straight line with a slope of $-\frac{K_a}{2.303}$



- Some drugs are absorbed very slowly. Other drugs are eliminated from the body very rapidly.
- In either case, absorption may be relatively slow compared to elimination and the absorption rate constant may be smaller than the elimination rate constant.
- In such cases, the slope of the linear segment of the semi log plot of C vs t is equal to $-\frac{K_a}{2.303}$ rather than $-\frac{K}{2.303}$ and the elimination rate constant must be determined by the method of residuals.

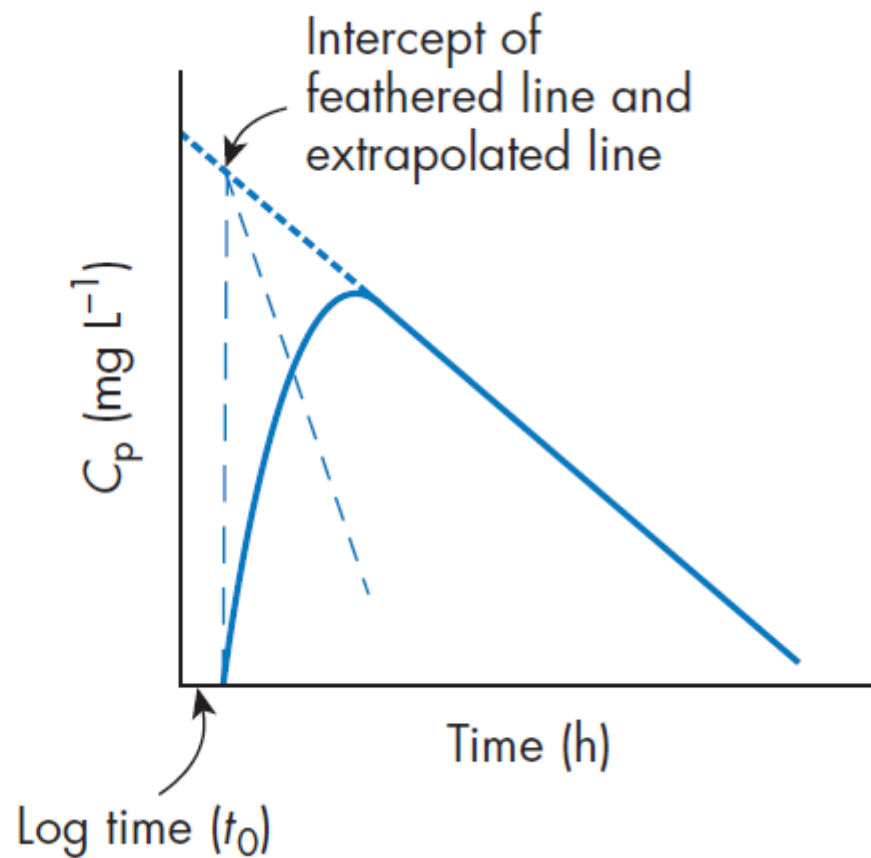
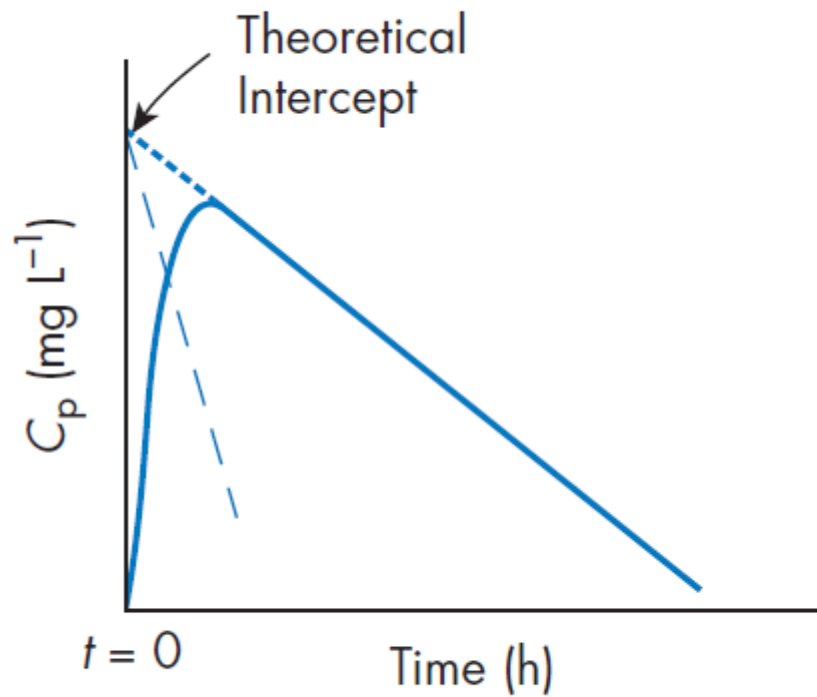


- Note:
 - In the method of residuals, the absorption process is assumed to be of the first order kinetics.
 - This assumption is valid for solutions and rapidly dissolving dosage forms, where the absorption process itself is rate determining.
 - In cases where the release of the drug from the dosage form is rate limiting, the kinetics are often zero order, mixed zero and first order, or even more complex processes.



Lag time (t_0)

- The time course of concentration in plasma of certain drugs suggests a time lag between oral administration and the apparent onset of absorption.
- This lag may be the result of delayed release of drugs from the dosage form or of a combination of negligible absorption from the stomach and slow gastric emptying.
- One usually concludes the existence of a 'lag time' if the intersection of the extrapolations of the terminal exponential phase and residual line occurs at a time greater than zero.





- When a lag is evident, the appropriate equation to describe the time course of drug concentrations in plasma is:

$$C = \frac{K_a F X_0}{V(K_a - K)} \left(e^{-K(t-t_0)} - e^{-K_a(t-t_0)} \right)$$



Determination of $C_{\max}$ and $t_{\max}$

- The peak time ($t_{\max}$) is the time at which the body displays the maximum plasma concentration ($C_{\max}$).
- It occurs when the rate of absorption is equal to the rate of elimination.
- At the peak time, therefore,

$$K_a(X_a)_{t_{\max}} = K(X)_{t_{\max}}$$



- According to the rate of change amount of the drug in body:

$$\frac{dX}{dt} = K_a X_a - KX$$

- At $t = t_{\max}$

$$\frac{dX}{dt} = K_a (X_a)_{t_{\max}} - K(X)_{t_{\max}} = 0$$

$$K_a (X_a)_{t_{\max}} = K(X)_{t_{\max}}$$



- The amount drug in the body at time t is given by:

$$(X)_t = \frac{K_a F X_0}{(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

- And, the amount to be absorbed:

$$(X_a)_t = F X_0 e^{-K_a t}$$



- Substituting $t_{\max}$ for t in the above equations

$$(X)_{t_{\max}} = \frac{K_a F X_o}{(K_a - K)} (e^{-K t_{\max}} - e^{-K_a t_{\max}})$$

- And,

$$(X_a)_{t_{\max}} = F X_o e^{-K_a t_{\max}}$$

- Then (substituting, rearranging and simplifying), we get

$$K_a e^{-K_a t_{\max}} = K e^{-K t_{\max}}$$

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- Rearranging the above equation:

$$\frac{K_a}{K} = \frac{e^{-Kt_{max}}}{e^{-K_a t_{max}}}$$

- Taking the logarithm of both sides and solving for t_{max} yields:

$$t_{max} = \frac{2.303}{K_a - K} \log \frac{K_a}{K}$$



- Maximum plasma concentration is described by substituting $t_{\max}$ for t in:

$$C_{\max} = \frac{K_a F X_0}{V(K_a - K)} (e^{-K t_{\max}} - e^{-K_a t_{\max}})$$

- But from $\frac{K_a}{K} = \frac{e^{-K t_{\max}}}{e^{-K_a t_{\max}}}$,

$$e^{-K_a t_{\max}} = \frac{K}{K_a} e^{-K t_{\max}}$$



- Therefore,

$$C_{max} = \frac{K_a F X_0}{V(K_a - K)} \frac{K_a - K}{K_a} e^{-K t_{max}}$$

- The above equation can be simplified to:

$$C_{max} = \frac{F X_0}{V} e^{-K t_{max}}$$



Urinary Excretion Data

- Read